



Self-Reflection: Choose your visualization type



Activity Overview

Now that you've been introduced to decision trees, pause for a moment and practice what you've been learning. In this self-reflection, you'll first review decision trees and data visualization best practices. Then, you'll examine a scenario, use a decision tree to determine the most effective chart type based on the scenario, and respond to brief questions.

This self-reflection will help you develop insights into your own learning and prepare you to use a decision tree to help you plan your next visualization. As you answer questions—and come up with questions of your own—you will consider concepts, practices, and principles to help refine your understanding and reinforce your learning. Answering and asking questions in this self-reflection will help to reinforce what you've learned so far, so it will be easier for you to remember it later.

Next, review decision tree uses, concepts, and best practices.

Decision tree uses

A **decision tree** is a flowchart that you can use to help frame larger decisions as a series of smaller yes/no decisions. These are useful when trying to choose the best data visualization to communicate a given message to your audience. Because different visualizations have different strengths and weaknesses, a decision tree can help you pick the best visualization for your data and audience. Here's the decision tree you examined earlier in this course:

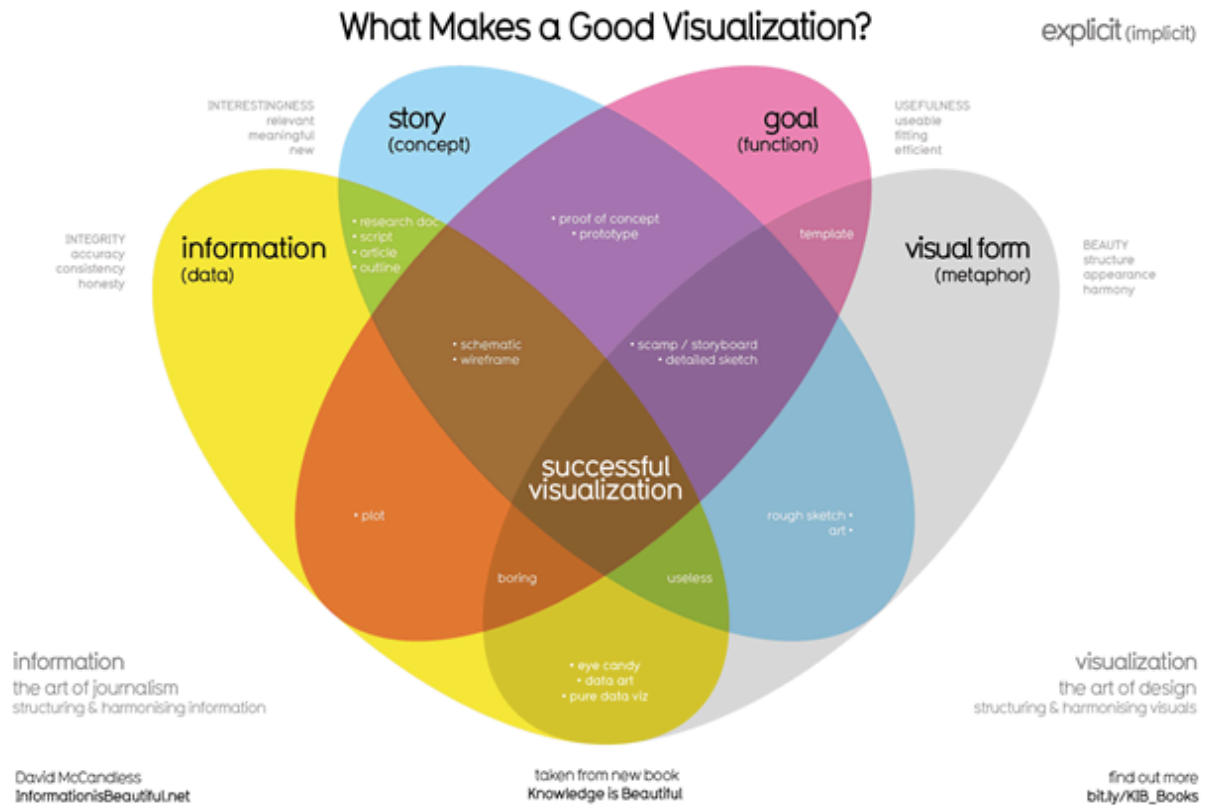


Path 1: Does your data have only one numeric variable? If so, choose a histogram or a density plot. Path 2: Are there multiple datasets? If so, choose a line chart or a pie chart. Path 3: Are you measuring changes over time? If so, choose a bar chart. Path 4: Do relationships between data need to be shown? If so, choose a scatterplot or a heatmap.

Decision tree concepts and best practices

Before you use the decision tree in a scenario to create successful data visualizations, take a moment to review some of the design concepts and best practices you've learned so far. Recall that there are four elements of successful data visualization:

- **Information:** reflects the conclusion you've drawn from the data, which you will communicate with visualization
- **Story:** adds meaning to the data and makes it interesting
- **Goals:** makes the data usable and useful
- **Visual form:** creates both beauty and structure



Keep these elements in mind as you review the scenario below. This will help you make better data visualizations by helping you connect the information you want to communicate with your audience and your goals.

Here are some additional best practices to keep in mind:

- Your audience should know what they are observing within five seconds of being shown a data visualization. Visuals should be clear and easy to follow.

- In the five seconds after that, your audience should understand the conclusion your visualization is making—even if they aren’t familiar with your research.
- As long as it’s not misleading, you should visually represent only the data that your audience needs to understand your findings. Including irrelevant data may confuse, distract, or overwhelm your audience.

These rules will guide you as you create your visualizations and you can apply them as you consider the following scenario.

You’re a junior data analyst at a local retailer. In your current data analysis project, you’ve been exploring sales data for all of your company’s products, including the top products and sales trends for the last year. You need to present the results of this analysis to the company executives. Your goal for this presentation is to demonstrate how sales of the company’s products have changed over the last 12 months. Your findings about the two top-selling products include:

Product name	Units sold	Time period with highest sales
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Product A	2.5 million	70% of total annual sales occur in October, November, and December
Product B	1.9 million	Mostly consistent sales year-round, with a slight increase in November, December, and January

Your audience is the chief marketing officer (CMO) and marketing department vice presidents (VPs), not other data analysts or engineers. The audience will use the information you present to make decisions on how to allocate the advertising budget for each product for the coming year.

Now, use the decision tree.

Consider what you know about the type of data you have (product sales data) and the kind of relationships you are trying to communicate (time periods with the most sales). Use this information to navigate the decision tree and use it to determine what type of visualization would be most appropriate.

Based on how you want to represent the sales data, there are several choices that would result in a successful chart. You could frame this by using a comparison of different categories with a bar graph or by showing how the composition of total sales changes month-to-month as a line chart with lines for each product.

Either of these would be a successful choice. Combined with effective use of design principles, either could accurately communicate the message you need to convey to company leadership. This message being: Sales for Product A are extremely seasonal, so they may want to consider reducing their advertising spend for this product when it's out of season.

1. Reflection

1 / 1 point

In your next project, you consider the sales of Product A across the country. Sales vary considerably by city within your country, and the executive team is interested in targeted advertising by region. Use the decision tree to determine which chart type is the best option for this project.

- ☒ Heatmap
- ☐ Line chart
- ☐ Histogram
- ☐ Scatterplot

✓ **Correct**

A heatmap is the best chart type because it shows differences in the sale of one product across cities in your country.

2. Consider the scenario and the decision tree you used in this reflection:

1 / 1 point

- How did a decision tree simplify your decision-making process?
- Did the result of the decision tree differ from your first instinct?

Now, write 2-3 sentences (40-60 words) in response to each of these questions. Type your response in the text box below.

Using a decision tree, I can decide what type of chart is most suitable for a particular type of case. For example, if there is a sale of a particular product quarterly, then I will use the bar chart or histogram.

Yeah, if we can't use an appropriate decision tree, then the result will differ, and the visualization will not be proper, and we won't get the proper insight about the data. So, the proper decision tree is necessary for choosing the proper chart.

Feedback

A successful reflection on this topic would include your first idea of which visualization to use, as well as the results of using the decision tree.

Decision trees allow you to break down big or overwhelming problems into smaller, more manageable decisions. By framing the decision as a flowchart of various conditions, constraints, and goals, you can approach it systematically. When selecting a data visualization, this tool can help you find the most appropriate visualization to clearly communicate a message about the data to your audience.

3. The decision tree presented in this activity can be used to determine chart type based on the type of data and the type of story you want to tell. How would you modify the decision tree to help you choose a chart type based on your audience?

1 / 1 point

Write 2-3 sentences (40-60 words) in response to this question. Type your response in the text box below.

If the data has one numerical value, then we have to choose a histogram or density plot. If there are multiple datasets, then we have to choose a pie chart or a line chart. If there was measuring over time, then we have to choose a bar chart. In the other case we have to choose a heatmap or a scatter plot.

Feedback

An effective response would consider that audiences may differ in their ability to interpret different types of charts. For example, you could present a more complex chart to an audience of data analysts than to a group of executives. Additionally, different audiences are likely to be interested in different metrics.

These considerations could be taken into account by creating a decision tree that starts with the question "What interests my audience?"